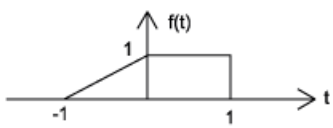
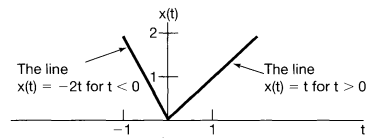
EED-201 (Signals and Systems): TUTORIAL 2

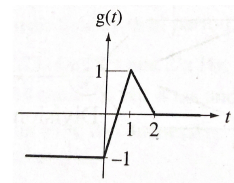
1. Find and sketch even and odd part of these signals:



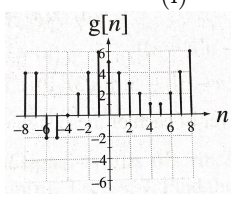
(i)



(ii)



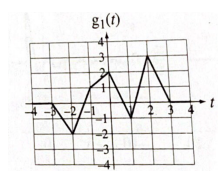
(iii)



(iv) v) $e^{-n/4}u[n]$ vi) $\sin(2\pi n/4)u[n]$

2. Determine whether the following signals are periodic or non-periodic. If periodic, find the fundamental time period.
   i) $2\cos(9\pi n/4) - 3\sin(6\pi n/5)$ ii) $e^{(-j6\pi n/21)} + \cos(22\pi n/36) - \sin(11\pi n/33)$
   iii) $\cos(2\pi t)u(t)$ iv) $e^{j[(n/4)-\pi]}$ v) $\cos(\pi n^2/8)$
   vi) $3\sin 20t + 8\cos 4t$ vii) $\cos^2 t$
   viii) $\cos 2\pi t + \sin 3\pi t + \cos(5\pi t - 3\pi/4)$.

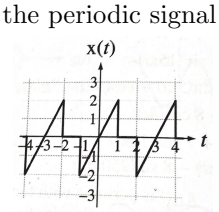
3. For the following figure, find and plot: i) $3g_1(2-t)$ ii) $-2g_1(t/4)$ iii) $g_1(0.5t-1.5)$.



4. A function $g[n]$ is defined by:

$$g[n] = \begin{cases} -2, n < -4 \\ n, -4 \leq n < 1 \\ 4/n, 1 \leq n \end{cases}$$

   Plot $g[-n],\ g[2-n],\ g[2n],\ g[n/2]$.

5. Determine whether the following signals are energy or power signals. Find the corresponding energy or power of the signals.
   i) $x[n] = u[n]$ ii) $x[n] = u[n] - 2u[n-4] + u[n-10]$ iii) $x[n] = -4\cos \pi n$
   iv)$x(t) = tu(t)$ v) $x(t) = \begin{cases} |t| - 1,\ |t| < 1 \\ 0,\ otherwise \end{cases}$

6. Find the average signal power of the periodic signal $x(t)$ shown below: